

Completing Two-Column Triangle-Congruence Proofs

Answer Key

High-school geometry

Quick Answers

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|---|---|
| 1. — | 6. — |
| 2. — | 7. — |
| 3. (b) $x = 5$, (c) the length = 17 cm, — | 8. (b) $x = 11$, (c) the measure = 59 degrees, — |
| 4. — | 9. — |
| 5. (b) $x = 12$, (c) the measure = 39 degrees, — | 10. (b) $x = 10$, (c) the length = 27 cm, — |
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What is verified

8 of 18 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: High-school geometry

HSG-CO.B – *problems 1, 3, 4, 5, 6, 7, 8, 10*

HSG-CO.C – *problems 2, 9*

Difficulty 1–5 of 5 across 18 tagged checks.

1. $\overline{AB} \cong \overline{CD}$, $\overline{BC} \cong \overline{DA}$; prove $\triangle ABC \cong \triangle CDA$. Fill in the reasons.

Reasons: 1. Given. 2. Given. 3. **Reflexive property of congruence** — the diagonal $\overline{AC}$ belongs to both triangles, so it is congruent to itself. 4. **SSS**.

Grading: line 3 must name the reflexive property; “it is the same side” is the observation, not the reason. Line 4 must name the criterion, not restate the conclusion.

2. $\overline{PX} \cong \overline{SX}$, $\overline{QX} \cong \overline{RX}$; prove $\triangle PQX \cong \triangle SRX$. Fill in the reasons.

Reasons: 1. Given. 2. **Vertical angles are congruent** — $\angle PXQ$ and $\angle SXR$ are the opposite pair formed by the two intersecting segments. 3. Given. 4. **SAS**.

Why SAS and not SSA: the angle sits *between* the two marked sides in each triangle, which is exactly the SAS pattern. Read the order down the table — side, angle, side — and the criterion reads itself off.

3. SAS given; $BC = 3x + 2$, $EF = 5x - 8$ cm. (a) Criterion and reason. (b) x . (c) BC .

(a) The three givens are a side, the angle *between* the two sides, and a second side, so the criterion is **SAS**. Once the triangles are congruent, $\overline{BC} \cong \overline{EF}$ follows by **CPCTC** — corresponding parts of congruent triangles are congruent. CPCTC is used after the criterion, never to establish it.

(b) CPCTC makes the two lengths equal:

$$3x + 2 = 5x - 8$$

$$10 = 2x$$

So $x = 5$.

(c) $BC = 3(5) + 2 = 15 + 2$. Check: $EF = 5(5) - 8 = 17$ too.

$x = 5$

$BC = 17$ cm

4. ASA; reasons printed, statements blank.

Statements:

$$1. \angle G \cong \angle J \quad 2. \overline{GH} \cong \overline{JK} \quad 3. \angle H \cong \angle K \quad 4. \triangle GHI \cong \triangle JKL$$

Why this order: the fourth reason is ASA, so the first three lines must be angle, included side, angle. $\overline{GH}$ is the side between $\angle G$ and $\angle H$, which is what makes the criterion ASA rather than AAS.

Grading: the three givens in any order earn most of the credit; naming the included side in the middle position earns all of it.

5. AAS given; $m\angle C = 4x - 9$, $m\angle F = 2x + 15$. (a) Criterion, and why not ASA. (b) x . (c) $m\angle C$.

(a) **AAS.** The marked side $\overline{BC}$ is not the side between $\angle A$ and $\angle B$ — it lies opposite $\angle A$ — and ASA requires the *included* side. AAS still proves congruence, because the third angle is forced by the angle sum, which turns the AAS pattern into an ASA one.

(b) By CPCTC, $\angle C \cong \angle F$:

$$4x - 9 = 2x + 15$$

$$2x = 24$$

So $x = 12$.

(c) $m\angle C = 4(12) - 9 = 48 - 9$. Check: $m\angle F = 2(12) + 15 = 39$ too.

$x = 12$

39°

6. HL; right angles at C and Z . Fill in the reasons.

Reasons: 1. Given. 2. **Definition of a right triangle** — a triangle with a right angle. 3. Given. 4. Given. 5. **HL**.

Why line 2 is not optional: the hypotenuse-leg criterion may only be cited once both triangles have been named right triangles. Without that line the proof is citing a rule whose hypothesis has not been established — and the same three parts in a non-right triangle would be side-side-angle, which proves nothing.

7. $\overline{AB} \cong \overline{AC}$, M the midpoint of $\overline{BC}$; reasons printed, statements blank.

Statements:

1. $\overline{AB} \cong \overline{AC}$ 2. M is the midpoint of $\overline{BC}$ 3. $\overline{BM} \cong \overline{MC}$ 4. $\overline{AM} \cong \overline{AM}$ 5. $\triangle ABM \cong \triangle ACM$

Reading the reasons backwards: line 3's reason is the definition of midpoint, so line 3 has to be the congruence that definition produces; line 4's reason is the reflexive property, so line 4 has to be the shared segment $\overline{AM}$. Three pairs of sides then give SSS.

8. SSS given; $m\angle B = 5x + 4$, $m\angle E = 3x + 26$. (a) Criterion and reason. (b) x . (c) $m\angle B$.

(a) Three pairs of congruent sides give **SSS**. The congruence of the angle pair then follows by **CPCTC**.

(b)

$$5x + 4 = 3x + 26$$

$$2x = 22$$

So $x = 11$.

$x = 11$

(c) $m\angle B = 5(11) + 4 = 55 + 4$. Check: $m\angle E = 3(11) + 26 = 59$ too.

59°

9. $\overline{PQ} \parallel \overline{SR}$ and $\overline{PQ} \cong \overline{SR}$; prove $\triangle PQX \cong \triangle SRX$. Fill in the reasons.

Reasons: 1. Given. 2. **Alternate interior angles** — $\overline{PQ} \parallel \overline{SR}$ cut by the transversal $\overline{PR}$ makes $\angle QPX$ and $\angle RSX$ alternate interior angles, hence congruent. 3. **Vertical angles are congruent**. 4. Given. 5. **AAS**.

Why AAS: the congruent side $\overline{PQ}$ lies opposite $\angle PXQ$ rather than between the two angle pairs, so the pattern is angle-angle-side. The parallel mark is doing real work here — it is the only thing that produces the first angle pair.

10. $\overline{AB} \cong \overline{CD}$, $\angle BAC \cong \angle DCA$; $BC = 2x + 7$, $DA = 4x - 13$ cm. (a) The whole proof. (b) x . (c) BC .

(a) **Model proof:**

1. $\overline{AB} \cong \overline{CD}$ — *Given*.
2. $\angle BAC \cong \angle DCA$ — *Given*.
3. $\overline{AC} \cong \overline{CA}$ — *Reflexive property of congruence*.
4. $\triangle BAC \cong \triangle DCA$ — *SAS*.

The criterion is SAS because in each triangle the marked angle sits between the marked side and the shared diagonal: $\angle BAC$ is between $\overline{AB}$ and $\overline{AC}$, and $\angle DCA$ is between $\overline{CD}$ and $\overline{CA}$.

(b) By CPCTC, $\overline{BC} \cong \overline{DA}$:

$$2x + 7 = 4x - 13$$

$$20 = 2x$$

So $x = 10$.

$x = 10$

(c) $BC = 2(10) + 7 = 20 + 7$. Check: $DA = 4(10) - 13 = 27$ too.

$BC = 27$ cm

Grading (a): four justified lines, with the reflexive property named for the diagonal and the criterion named on the last line. Any order that keeps every line justified is acceptable.